In this paper, every Ring is Commutative Ring with $1 \neq 0$.

「Thm. Let $A_1, \cdots, A_k$ be Ideals in R. Define a map

$$\varphi : R \to R/A_1 \times \cdots \times R/A_k : r \mapsto (r+A_1, \cdots, r+A_k)$$

is a Ring Homomorphism with $\ker \varphi = A_1 \cap A_2 \cap \cdots \cap A_k$.
If for each $1 \le i < j \le k$, $A_i + A_j = R$ (comaximal), then φ is onto and

$$\ker \varphi = A_1 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k.$$

Therefore, $R/(A_1 \cdots A_k) = R/(A_1 \cap \cdots \cap A_k) \cong R/A_1 \times \cdots \times R/A_k$.
Proof. First, prove this for $k=2$. The map

$$\varphi : R \to R/A_1 \times R/A_2 : r \mapsto (r+A_1, r+A_2)$$

is just the natural projection for R/A_1, R/A_2 elementwise, so ring-Homomorphism.
Moreover,

$$\varphi(r) = 0 \iff r+A_1 = A_1 \text{ and } r+A_2 = A_2$$
$$\iff r \in A_1 \text{ and } r \in A_2$$

So $\ker \varphi = A_1 \cap A_2$ Now, suppose that $A_1 + A_2 = R$. Since $1 \in R = A_1 + A_2$,

$$x + y = 1 \text{ for some } x \in A_1 \text{ and } y \in A_2.$$

Therefore, $\varphi(x) = \varphi(1-y) = (x+A_1, 1-y+A_2) = (0+A_1, 1+A_2)$.
And similarly $\varphi(y) = (1, 0)$. Now, given $(r_1 + A_1, r_2 + A_2) \in R/A_1 \times R/A_2$,

$$\begin{aligned}\varphi(r_2 x + r_1 y) &= \varphi(r_2)\varphi(x) + \varphi(r_1)\varphi(y) \\ &= \varphi(r_2)(\bar{0}, \bar{1}) + \varphi(r_2)(\bar{1}, \bar{0}) \\ &= (\bar{0}, \bar{r_2}) + (\bar{r_1}, \bar{0}) = (r_1 + A_1, r_2 + A_2).\end{aligned}$$

Thus φ is onto. And $A_1 A_2 \subseteq A_1 \cap A_2$, because: for $a_i \in A_1$, $b_i \in A_2$,

$$a_1 b_1 + \cdots + a_n b_n \in A_1 \text{ and } A_2 \text{ since } A_1, A_2 \text{ are Ideal.}$$

Meanwhile, if $A_1 + A_2 = R$, for any $c \in A_1 + A_2$.

$$c = c \cdot 1 = c(x+y) = cx + cy \in A_1 + A_2.$$

So Proved(?)

Thm. Let R be a ring with identity $1 \neq 0$.
Let $E \in R$ be an idempotent in R s.t. $Er = rE$ for all $r \in R$.
Then, $RE = R(1-E)$ are two-sided ideals of R, and that

$$R \cong RE \times R(1-E).$$

Moreover, E and $1-E$ are identities for RE and $R(1-E)$ respectively.

Proof.

$$DE_1 \cdots E_n =$$